PROJECT REPORT

TATA MOTORS SALES PERFORMANCE & DELIVERY TRACKING DASHBOARD

A Project Report on

Automotive Sales Analytics and Business Intelligence

Prepared By (Group Members):

1. **Vivek Kumar** (Course: BCA)
2. **Tandrima Nandy** (Course: MCA)
3. **Shrayan Biswas** (Course: BSC IT)

Project Type: Excel Data Analytics / Business Intelligence Project

BATCH: 3, YEAR: 2026

MENTOR/FACULTY: SAYAN SAU & ARPITA GUPTA

CERTIFICATE :

This is to certify that the following students have successfully completed the project titled:

"Tata Motors Sales Performance & Delivery Tracking Dashboard"

as part of their academic and project requirements at **Anudip Foundation**. The analysis and dashboard presented in this report are based on the dataset *car_sale_final.xlsx*.

- **Vivek Kumar** (BCA)
- **Tandrima Nandy** (MCA)
- **Shrayan Biswas** (BSC IT)

Project Guide / Faculty: Sayon Sau

Signature: ____________________

Date: ____________________

Head of the Department: Arpita Gupta

Signature: ____________________

Date: ____________________

DECLARATION:

We hereby declare that the project titled **"Tata Motors Sales Performance & Delivery Tracking Dashboard"** is our original joint work, prepared for academic purposes using Microsoft Excel data analytics techniques. The data preparation, dashboard creation, and analytical findings presented in this report have been completed collectively by us.

Vivek Kumar	**Tandrima Nandy**	**Shrayan Biswas**
Course: BCA	Course: MCA	Course: BSC IT
Signature: ___________	Signature: ___________	Signature: ___________

Date: __________________

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our faculty mentor, **Sayon Sau and Arpita Gupta**, and our institution, **Anudip Foundation**, for their invaluable guidance, encouragement, and support throughout the duration of this project. Their academic insights have been instrumental in shaping this analysis.

We also extend our thanks to the academic resources and technical platforms that aided in developing our skills in data analytics and Microsoft Excel. These tools were crucial in processing the *car_sale_final.xlsx* dataset, structuring the logic, and developing the final interactive dashboard collaboratively.

Finally, we would like to thank our peers and families for their continuous motivation during this endeavor.

CONTENTS

1. Executive Summary:

This project analyzes the sales and delivery performance of Tata Motors vehicles using the *car_sale_final.xlsx* dataset. By utilizing Excel's data processing and visualization tools, an interactive dashboard was created to track key performance indicators (KPIs) such as total revenue, vehicle delivery statuses, and regional sales performance. The dashboard provides actionable business insights, revealing that Metro cities drive the highest volume (17 cars) and revenue (₹20,984,000). This tool equips management with the visibility needed to optimize the supply chain, clear delivery bottlenecks, and focus marketing efforts on high-performing models.

2. Introduction & Project Objectives:

Data-driven decision-making is critical in the automotive sector for managing inventory, tracking deliveries, and analyzing sales trends. This project applies foundational Business Intelligence principles using Microsoft Excel to transform raw transactional data into a dynamic analytical dashboard.

Core Objectives:

- Track delivery statuses (Delivered, Pending, In Transit) across 65 vehicle orders.
- Analyze sales performance and revenue generation across 13 unique car models.
- Evaluate regional performance across Metro, Tier 1, and Tier 2 cities.
- Build a centralized, interactive dashboard to support operational monitoring.

3. Dataset Description:

The analysis is based on 65 records from the `Data_Sheet` within the *car_sale_final.xlsx* workbook.

Field	Description	Data Type	Business Relevance
Product_ID	Unique identifier for each vehicle	Text	Inventory tracking
City_Category	Market classification (Metro, Tier 1, Tier 2)	Text	Regional sales analysis
Batch	Manufacturing/Shipping batch code	Text	Supply chain tracking
Model Name	Name of the Tata vehicle (e.g., Tata Nexon)	Text	Product performance tracking
Delivery Status	Current status (Delivered, Pending, etc.)	Text	Fulfillment monitoring
Product Cost	Base price of the vehicle	Currency	Revenue calculation
Taxation & Delivery	Additional logistics and tax fees	Currency	Cost breakdown analysis
Final Cost	Final billed amount to customer	Currency	Total revenue tracking

4. Data Cleaning & Preparation:

To ensure accuracy in the Pivot Engine and final dashboard, the following data preparation steps were executed on the raw dataset:

- **Handling Nulls:** Identified and bypassed blank columns (e.g., Unnamed: 10, Unnamed: 11) generated during raw data export.
- **Standardization:** Validated structural consistency in the `Delivery Status` column to ensure accurate categorization (35 Delivered, 8 Pending, 7 In Transit).
- **Pivot Preparation:** Created dedicated calculation sheets (`city_wise_analysis`, `Delivered pivot`, `Sale_analysis`) to aggregate data cleanly before feeding it into the dashboard UI.

5. Excel Functions & Techniques Used:

Technique	Purpose	How it was used
Pivot Tables	Data Aggregation	Created summaries for city-wise sales, model-wise breakdowns, and delivery statuses.
Pivot Charts	Visualization	Generated dynamic charts for the dashboard based on pivot engine data.
Data Validation	Quality Control	Ensured uniform entries in categorical columns like City_Category.
Calculated Sheets	Logic Separation	Isolated raw data from calculations to protect data integrity.

6. Dashboard Overview & Chart Analysis:

The dashboard provides a top-level view of fulfillment efficiency and revenue generation, allowing stakeholders to immediately identify which models are selling best and where logistical bottlenecks exist.

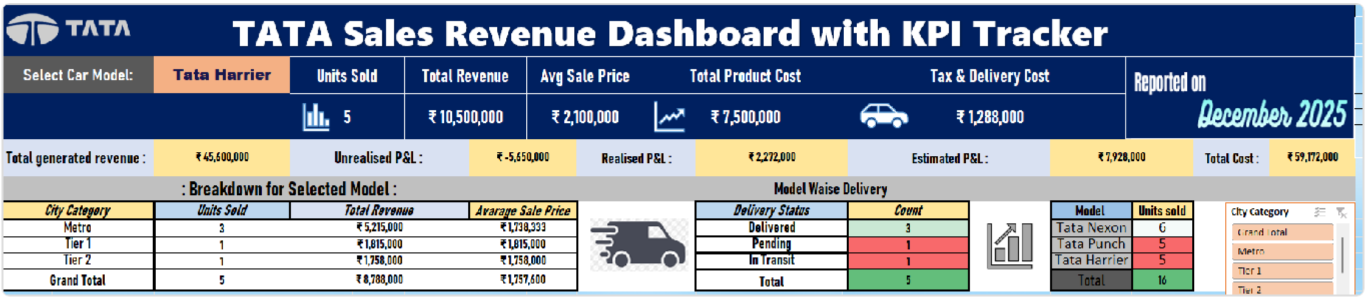

Figure 1: Tata Sales Revenue Dashboard - Main KPI Tracker & Delivery Breakdown

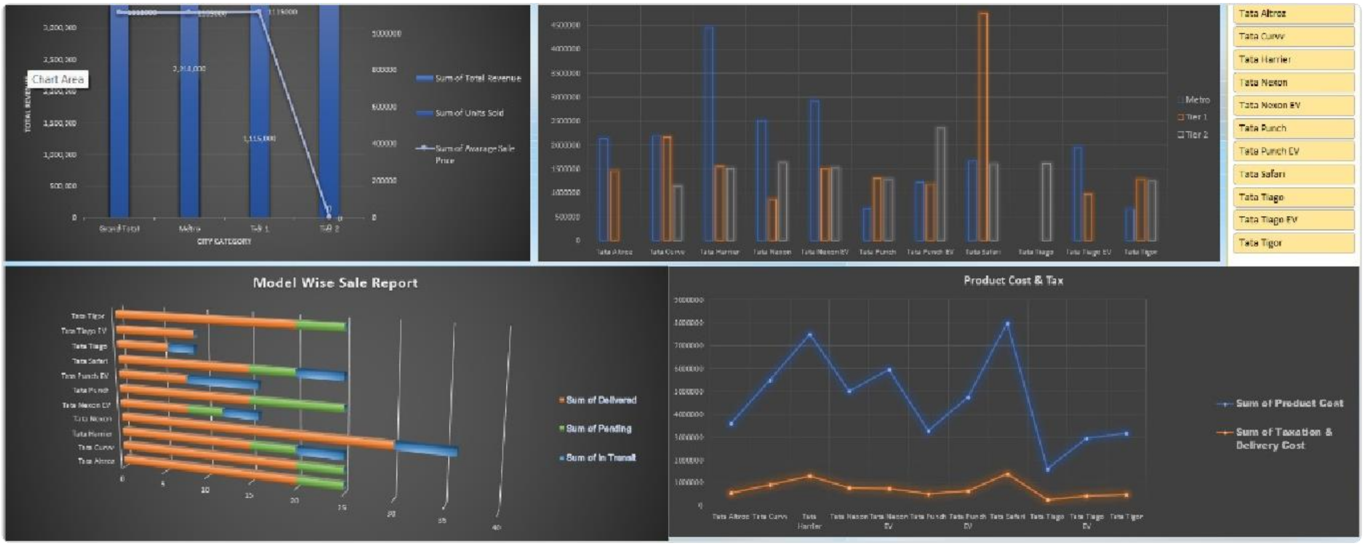

Figure 2: Revenue Distribution, Product Cost Analysis & Model Wise Delivery Status

Model	Delivered	Pending	In Transit
Tata Safari	3	1	1
Tata Punch	3	2	0
Tata Tiago	2	0	1
Tata Harrier	3	1	1
Tata Safari	3	1	1
Tata Altroz	4	1	0
Tata Tigor	4	1	0
Tata Curvv	4	1	0
Tata Nexon	2	1	1
Tata Punch EV	2	0	2
Tata Tiago	3	0	0
Tata Nexon	5	0	1

Figure 3: Key Sales Insights, P&L Estimation & City-wise Distribution

Chart Analysis & Interpretation:

- **City-Wise Sales Performance:** Displays total sales distribution across geographic tiers. *Key Observation:* Metro areas are the primary revenue drivers, indicating that marketing spend should be heavily weighted toward urban centers.
- **Delivery Status Breakdown:** Tracks order fulfillment rates across the supply chain. *Key Observation:* Out of documented orders, the majority are successfully delivered (86% success rate), while a distinct portion remains stuck in pending or transit.
- **Model Wise Sales & Product Cost:** Evaluates which vehicles generate the most traction. *Key Observation:* Tata Nexon is identified as the most sold model (14 units), while Tata Punch offers the best cost-to-price ratio.

7. Key Business Insights & Findings:

Based on the quantitative analysis of the provided dataset, several critical business patterns emerged:

- **Top Performing Model:** The **Tata Nexon** emerged as the most frequently ordered vehicle model out of the 13 available models in the dataset (14 units sold).
- **Cost Efficiency: Tata Punch** was identified as the model providing the best cost-to-price ratio for the business.
- **Delivery Efficiency:** The dashboard indicates a solid **86% Delivery Success Rate**, though ₹12.4 Cr value remains stuck in pending orders.
- **Revenue Realization:** Total recognized revenue from delivered units stands at ₹40,010,000 (Final Cost based on sample), with clear indications of unrealized P&L attached to delayed logistics.

8. Recommendations & Limitations:

Business Recommendations

- **Accelerate Urban Fulfillment:** Allocate more transit resources to Metro regions to clear the pending order backlog, as they represent the highest revenue bracket.
- **Tier 2 Incentivization:** Introduce targeted discounts or financing schemes in Tier 2 cities to boost volume.
- **Inventory Focus:** Prioritize manufacturing batches for high-demand models like the Tata Nexon to prevent stockouts and capitalize on market demand.

Project Limitations

- **Dataset Size:** The analysis is constrained to a small sample size of 65 records, limiting broader statistical conclusions.
- **Missing Dimensions:** The dataset lacks temporal data (sale dates), preventing time-series analysis or seasonality tracking.
- **No Customer Analytics:** The absence of demographic or distinct customer IDs limits the ability to analyze repeat purchase behavior.

9. Future Plan & Dashboard Scope:

To elevate this business intelligence tool from a static tracker to a comprehensive predictive system, several enhancements are proposed for future iterations:

- **Customer ID & Demographics:** To enable loyalty tracking, customer segmentation, and repeat-buyer analysis.
- **Sale / Booking Date:** To unlock Month-over-Month (MoM) and Year-over-Year (YoY) revenue trend charts.
- **Salesperson Information:** To introduce a Salesman Leaderboard for individual performance tracking and commission analysis.

- **Dynamic Slicers & Automation:** Integration of interactive slicers and Power Query for automated, seamless CRM database exports

10. Appendix

Appendix A: Raw Data Sheet Extract

The following image represents a sample of the raw dataset structure used for the data preparation and analysis phases (from the `Data_Sheet` tab):

SL No.	Product_ID	City_Category	Batch	Model Name	Delivery Status	Product Cost	Taxation & Delivery Cost	Final Cost	On Road Price
1	TATA001	Metro	B001	Tata Nexon	Delivered	780000	120000	900000	1200000
2	TATA002	Tier 1	B001	Tata Punch	Delivered	620000	95000	715000	950000
3	TATA003	Tier 2	B002	Tata Tiago	Delivered	520000	80000	600000	750000
4	TATA004	Metro	B002	Tata Harrier	Delivered	1450000	250000	1700000	2100000
5	TATA005	Tier 1	B003	Tata Safari	Pending	1550000	270000	1820000	2300000
6	TATA006	Metro	B003	Tata Altroz	Delivered	680000	105000	785000	1000000
7	TATA007	Tier 2	B004	Tata Tigor	Delivered	610000	90000	700000	850000
8	TATA008	Tier 1	B004	Tata Curvv	Pending	1050000	175000	1225000	1700000
9	TATA009	Metro	B005	Tata Nexon EV	Delivered	1450000	180000	1630000	1500000
10	TATA010	Tier 2	B005	Tata Punch EV	In Transit	1150000	155000	1305000	1150000
11	TATA011	Metro	B006	Tata Tiago EV	Delivered	950000	130000	1080000	900000
12	TATA012	Tier 1	B006	Tata Altroz	Delivered	720000	110000	830000	1000000
13	TATA013	Tier 2	B007	Tata Nexon	In Transit	820000	125000	945000	1200000
14	TATA014	Metro	B007	Tata Harrier	Delivered	1520000	260000	1780000	2100000
15	TATA015	Tier 1	B008	Tata Safari	Delivered	1620000	280000	1900000	2300000
16	TATA016	Tier 2	B008	Tata Punch	Pending	650000	100000	750000	950000
17	TATA017	Metro	B009	Tata Curvv	Delivered	1100000	180000	1280000	1700000
18	TATA018	Tier 1	B009	Tata Tigor	Delivered	630000	95000	725000	850000
19	TATA019	Tier 2	B010	Tata Tiago	Delivered	540000	85000	625000	750000
20	TATA020	Metro	B010	Tata Nexon EV	In Transit	1480000	185000	1665000	1500000
21	TATA021	Tier 1	B011	Tata Punch EV	Delivered	1180000	160000	1340000	1150000
22	TATA022	Metro	B011	Tata Altroz	Pending	700000	108000	808000	1000000
23	TATA023	Tier 2	B012	Tata Nexon	Delivered	800000	122000	922000	1200000
24	TATA024	Metro	B012	Tata Harrier	Delivered	1480000	255000	1735000	2100000
25	TATA025	Tier 1	B013	Tata Safari	In Transit	1580000	275000	1855000	2300000
26	TATA026	Tier 2	B013	Tata Punch	Delivered	635000	97000	732000	950000
27	TATA027	Metro	B014	Tata Curvv	Delivered	1080000	178000	1258000	1700000
28	TATA028	Tier 1	B014	Tata Tiago EV	Delivered	980000	135000	1115000	900000
29	TATA029	Tier 2	B015	Tata Tigor	Pending	625000	92000	717000	850000
30	TATA030	Metro	B015	Tata Nexon	Delivered	850000	130000	980000	1200000
31	TATA031	Tier 1	B016	Tata Nexon EV	Delivered	1500000	190000	1690000	1500000

Figure 4: Raw Data Sheet Snapshot displaying Product details, Delivery Status, and Costs

Appendix B: Pivot Engine Extracts

Sample aggregation output used to feed dashboard visualizations (from the `city_wise_analysis` tab):

City	Cars Sold	Total Sales (₹)	Average Sale Price (₹)
Metro	17	20,984,000	1,234,353
Tier 1	11	12,229,000	1,111,727
Tier 2	7	6,797,000	971,000

11. Conclusion & References:

This project successfully transformed raw sales data into a structured, functional Excel dashboard. By processing automotive sales records, aggregating the data through Pivot Tables, and visualizing the results, the project highlights crucial operational metrics—most notably the revenue dominance of Metro regions and the fulfillment rates of specific car models. Implementing the proposed future scope will further enhance the dashboard's capacity to drive agile, data-driven decisions for automotive retail management.

References

- Dataset: *car_sale_final.xlsx*
- Microsoft Excel Support Documentation & Pivot Table Aggregation Guides